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Elevated Convection

This is a sample lesson page from the [Certificate of Achievement in Weather Forecasting](#) offered by the Penn State Department of Meteorology. Any questions about this program can be directed to: [Steve Seman](#)

Prioritize...

Upon completion of this page, you should be able to discuss common synoptic-set ups that can lead to elevated convection, be able to recognize favorable environmental profiles for elevated convection on skew-T diagrams, and discuss the consequences for severe weather threats from elevated convection (compared to surface-based convection).

Read...

In the last section, you learned about the basic difference between elevated and surface-based thunderstorms, namely that an elevated thunderstorm is a type of deep, moist convection with an updraft originating above the planetary boundary layer. In contrast, updrafts associated with surface-based convection originate at the ground. As a reminder, elevated convection develops above a stable layer of air next to the ground (above a nocturnal inversion or above a stable layer on the cool side of anafronts). The environmental temperature profiles that produce elevated thunderstorms also look quite a bit different from those that produce surface-based thunderstorms, as these [idealized skew-T diagrams](#) from the last section illustrate.

The most common synoptic-scale set up that results in elevated thunderstorms occurs on the cool side of anafronts, where warm-air advection and upper-level divergence can work in tandem to initiate elevated convection. With these two processes in mind, elevated convection forms where unstable parcels of air [feed into the "bottom" of updrafts along the sloping frontal boundary](#).

While you've seen that general background before, I think the photograph below will really help you conceptualize how elevated convection "works" (a picture is worth a thousand words, right?). The photographer who captured this "striking" photograph had a high vantage point overlooking the Alexander Valley near Santa Rosa, California. At the time, stratus clouds shrouded the valley, indicating a stable layer of air in the lowest levels of the troposphere. Yet, above this stable layer, a bona fide thunderstorm was able to develop--a classic case of elevated convection. If you picture an elevated thunderstorm over stratiform clouds (a stable layer), you'll never confuse elevated convection with surface-based convection.

Photograph of an elevated thunderstorm over a stable layer in the Alexander Valley near Santa Rosa, California.

An elevated thunderstorm over a stable layer in the Alexander Valley near Santa Rosa, California.
Credit: Kent Porter

With your conceptual understanding of elevated convection hopefully cemented, now it's time to dial in on the synoptic patterns and thermal profiles that favor elevated convection. For starters, think back to the cluster of elevated thunderstorms from July 14, 2010 that we discussed previously. Without reservation, the synoptic-scale pattern at the time fit the definition of elevated convection "to a tee" because the thunderstorms formed north of a low-pressure system's warm front ([14Z radar reflectivity](#); [12Z surface analysis](#)). More to the point, [warm-air advection](#) and upper-level divergence (associated with [vorticity maxima ahead of a closed 500-mb low](#)) paved the way for elevated convection to form along the sloping frontal boundary.

We have yet to examine temperature soundings in the vicinity of these thunderstorms that formed on the morning of July 14, 2010. For sake of argument, let's focus our attention on International Falls, Minnesota, where a thunderstorm occurred at 14Z (check out [KINL's meteogram](#)). Below is the 14Z model analysis sounding at International Falls. First, note the deep, saturated layer (relative humidity essentially equal to 100%), which is

the skew-T footprint of precipitation. Second, note the strong stability in the layer of air from about 940 mb to roughly 800 mb (this layer of relatively warm air is consistent with warm-air advection north of the warm front). For all practical purposes, unstable parcels of air feeding into the "bottom" of the thunderstorm's updraft originated near 800 mb.

14Z sounding for KINL showing a set up for elevated convection.

The 14Z (model analysis) temperature and dew-point soundings at International Falls, MN, on July 14, 2010. At the time, unstable parcels of air fed into the bottom of the updraft of an elevated thunderstorm at the top of a stable layer (a layer of relatively warm air) between 940 mb and 800 mb.

Credit: Penn State University

Any thunderstorm that developed in this environment had to be elevated because a parcel lifted from the surface would *never* be positively buoyant (it would always remain to the left of the temperature sounding). To drive home this point, please open this [interactive tool](#) demonstrating why cool, stable, saturated air near the surface only allows for elevated convection (keep the tool open as you read through the instructions in the following paragraph).

Start by clicking on the red indicator on the right and slowly decrease the pressure level (increase the altitude) of a test air parcel. As you raise the red indicator, you'll see the parcel's moist adiabat (in dark blue). Clearly, the temperature of the air parcel at any point along the moist adiabat is lower than the environment's temperature. In short, the test parcel is negatively buoyant through a deep layer all the way up to pressures approaching 750 mb (the top of the inversion). A test air parcel nudged upward at the top of the temperature inversion would obviously be positively buoyant (note that the local moist adiabat switches from blue to orange at the top of the inversion).

So, the strong temperature inversion starting around 925 mb puts a tight lid on any parcels trying to rise relatively far above the earth's surface. Only air parcels located atop the temperature inversion could rise, in concert with warm-air advection and upper-level divergence to give them an initial boost. So, there's no way convection could be surface-based in this case. Although the inversion is much more dramatic in the interactive tool, the same pattern holds true in the skew-T for International Falls above. Only elevated convection is possible.

Even though CAPE tends to be smaller in elevated convection situations, elevated thunderstorms can still be severe. In most cases, the greatest threat from severe elevated convection is large hail. Damaging winds and tornadoes are still possible with *some* elevated thunderstorms, but these threats are greatly reduced, depending on the depth of the stable layer of air in the lower troposphere, below the altitude where unstable air parcels feed into the storm's updraft.

Indeed, the speed of downdrafts in elevated storms is often [no match for the depth and strength of the stable layer next to the ground](#). In other words, this stable layer discourages downdrafts from splashing down to earth because air parcels in downdrafts become warmer than their immediate surroundings. Thus, the threat of damaging straight-line winds from elevated storms is rather limited, as is the threat from tornadoes (we'll get into the reasons why stability in the lower troposphere discourages tornadogenesis later). Only when the stable layer is very thin (and stability in the layer is weak) can downdrafts occasionally penetrate to the surface, paving the way for damaging wind gusts, or a rare tornado.

Experienced mesoscale forecasters, like those at SPC, apply this conceptual understanding about the differing severe threats from surface-based and elevated storms regularly in their forecasting routine. For example, on the morning of August 19, 2005, elevated thunderstorms developed across Nebraska, and it's not hard to see why they were elevated based on the skew-T diagram from North Platte, Nebraska, on the right below (note the low-level inversion on the sounding).

 **(Left): 12Z August 19, 2005 radar mosaic showing clusters of thunderstorms over Nebraska. (Right): Corresponding 12Z sounding from North Platte, Nebraska**

(Left) The 12Z composite radar image (large) over the Central Plains on August 19, 2005, shows clusters of thunderstorms over Nebraska. (Right) The 12Z sounding at North Platte, Nebraska, indicates a low-level inversion. For the record, the pressure level at the top of the inversion was roughly 880 mb. Although elevated convection was not occurring at North Platte at this time, this skew-T served as a proxy for the environment in which elevated convection developed.
Credit: WSI / University of Wyoming

On the morning of August 19, a [stationary front was draped across Kansas](#). Nebraska's location on the cool side of an anafront was a "big picture" sign that the convection across the state would be elevated (confirmed by the environment depicted on the skew-T). At 1555Z, forecasters at SPC issued [Mesoscale Discussion #2029](#), which clearly indicated that the initial risk from elevated storms was large hail (and heavy rain). However, as storms developed farther south over Kansas later in the day, forecasters expected storms would become surface-based, increasing the threat of damaging straight-line winds (a couple of tornadoes actually ended up occurring, too).

Elevated convection can be an issue during the cold season, too, and in fact, it helps to explain some of the "weird" winter observations you sometimes hear about or experience: thunder with freezing rain, sleet, or snow. I should note that not all instances of thunder snow, in particular, are a result of elevated convection. Nevertheless, wintry precipitation falls if the environment favors elevated convection and the temperature sounding is entirely below 0 degrees Celsius (a classic snow sounding) or if the lower half of the troposphere makes a warm-air sandwich (a sleet or freezing-rain sounding). Let's explore the topic of wintry precipitation accompanied by thunder by taking a look at a *Case Study* about one of the worst ice storms ever to occur in Oklahoma.

Case Study...

Thunder with Freezing Rain

Photograph of an ice sculpture removed from the top of a fire hydrant in Norman, Oklahoma

A photograph of an ice sculpture removed from the top of a fire hydrant in Norman, Oklahoma, during the aftermath of a major ice storm on December 8-11, 2007.

Credit: Greg Carbin

On December 8-11, 2007, an ice storm crippled parts of Oklahoma and other Midwestern States. Freezing rain produced as much as three inches of ice from Oklahoma City to Tulsa, bringing down trees and power lines. The unique photograph on the right shows an "ice sculpture" after it was carefully removed from the top of a fire hydrant in Norman, Oklahoma, near the end of this major ice storm. As heavy freezing rain fell occasionally during this storm, lightning flashed in the clouds above, so this case gives us a dramatic example of cold-season elevated thunderstorms!

The 08Z sounding at Tulsa Oklahoma, on December 10 (below) shows the classic juxtaposition of the relatively warm, moist layer near 850 mb and the shallow Arctic air mass near the ground. Recall that this warm-air sandwich (a slice of warm air between cold air at higher altitudes and cold air near the surface) is a classic recipe for freezing rain. With 850-mb dew points rather high, the National Weather Service called for heavy freezing rain, which, of course, eventually verified (check out the [meteogram for Tulsa, OK, on December 10](#)). By the way, did you notice the symbols for lightning on the meteogram?

08Z skew-T for Tulsa, OK on December 10, 2007, showing a classic freezing rain sounding.

The 08Z sounding at Tulsa, OK, on December 10, 2007, reveals a classic freezing rain sounding. At this time, the weather observer recorded lightning (but no thunder), indicating that the flashes were associated with an elevated thunderstorm. Indeed, there was no way convection originated at the ground because the lower troposphere was so stable.

Credit: Penn State University

The strong temperature inversion in the lower troposphere on the Tulsa sounding means that there was no way convection originated at the ground. You may be wondering how forecasters assess the potential for strong

updrafts in situations like this. After all, CAPE for a surface parcel here would be zero. As it turns out, however, we have alternative methods of calculating CAPE, which can better catch the potential for strong updrafts in elevated convection situations.

We'll explore alternative versions of CAPE more later on, but for now check out the [08Z field of lapse rates between 700 mb and 500 mb](#). Focus your attention on northeast Oklahoma, where lapse rates were between 6.5 and 7 degrees Celsius per kilometer. Although the moist adiabatic lapse rate is variable, we can use 6 degrees Celsius per kilometer as a representative value. With this threshold in mind, you can see that, over northeast Oklahoma, the layer between 700 mb and 500 mb was unstable with respect to moist ascent at this time. Parcels rising from just above the top of the inversion, which were already saturated (or very nearly so), would be positively buoyant if nudged upward, so the middle troposphere supported elevated convection.

How did this atmospheric profile, which supported freezing rain and elevated thunderstorms, come to be? Let's take a look at the synoptic-scale weather pattern that set the stage for the ice storm in Oklahoma (December 8-11, 2007). This big-picture assessment will give you a better sense for how elevated thunderstorms developed on this day and, as a result, will hopefully provide you with insights that will help you to forecast elevated convection.

The Synoptic-Scale Set-Up for the Oklahoma Ice Storm...

The weather pattern favorable for an ice storm (with lightning!) in Oklahoma and surrounding states displayed the characteristics you might expect when dealing with elevated convection:

1. warm advection (overrunning) on the cold side of an anafront
2. a source of upper-level divergence to help nudge parcels upward
3. fairly steep mid-level lapse rates

Add an Arctic air mass associated with a strong area of high pressure, and you have the low-level chill needed for ice! For starters, the [00Z surface analysis on December 10, 2007](#) showed the Arctic air mass spilling southward over the southern Plains (the double-barreled high over the Plains and Upper Midwest marked the center of the air mass). Meanwhile, the flow of air at 850 mb had turned southerly around a center of high 850-mb heights over the Southeast States. As a result of this flow aloft back across the cold front (this was an anafrontal cold front), the stage was set for overrunning precipitation in the form of freezing rain. The [00Z analysis of 850-mb temperature advection](#) confirms another classic warm-air advection pattern for elevated convection.

Granted, warm advection at 00Z on December 10 was relatively weak, but SPC forecasters warned that a developing low-level jet stream would eventually enhance warm advection and cause freezing rain to persist over the region. You can see the footprint of the developing low-level jet stream on the analysis of 850-mb isotachs and wind barbs at 00Z on December 10 below.

850-mb wind barbs and isotachs at 00Z, December 10, 2007.

A model analysis of 00Z wind barbs and isotachs at 850 mb, on December 10, 2007. The footprint of the developing low-level jet stream was evident (wind speeds greater than 30 knots) over northeast Texas toward the Middle Mississippi Valley.

Credit: Penn State University

In time, the deep moisture convergence associated with the low-level jet stream expanded over Oklahoma (check out the [04Z analysis on December 10](#), for example), as the low-level jet stream became more established over Oklahoma. This set up alone would have given us an ice storm, given the Arctic chill near the surface, and the temperature profile on the Tulsa sounding above. But, with some upper-level divergence to give a boost to parcels above the cold, stable surface layer so that they could continue rising via their own positive buoyancy, the stage was set for "thunder ice."

In this particular case, the upper-level divergence came from a mid-level wind maximum, which forecasters at SPC referred to in [Mesoscale Discussion #2203](#) (among others). The [00Z model analysis of 500-mb winds](#) shows that Oklahoma and other ice-affected areas lay in the right entrance of a mid-level jet streak. In case you're wondering, the dynamics of mid-level speed maxima like this one are similar to straight 300-mb jet streaks (divergence in the right-entrance and left-exit regions).

At any rate, the ingredients for heavy freezing rain occasionally punctuated by lightning came together for a crippling and memorable ice storm over Oklahoma and parts of the surrounding states. There are some awesome [photographs of the ice storm](#) on the Web site of the National Weather Service in Norman. I hope that you now have a better appreciation for how the synoptic-scale weather pattern plays a pivotal role in paving the way for elevated convection. In case you're wondering, anafrontal cold fronts aren't just a player in elevated convection situations during the winter. If you're interested in seeing a case of how an anafrontal cold front helped spawn elevated convection above a frontal inversion in the warm season, check out this [Explore Further video](#) ([video transcript](#)).

Throughout this discussion of how the big picture relates to mesoscale weather, I have yet to mention 300 mb. Lest I leave you with the impression that this lofty pressure level is unimportant in the grand scheme of elevated convection (or the topic of severe weather), we'll turn our attention to the top of the troposphere in the next section.

[◀ The Big Picture at 850 mb Upper-Level Jet Streaks ▶](#)



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